

## Section 4 Random Variables

Statistics I · Fall 2026 · Zong-Wei Yeh · Department of Finance, CCU

### Overview

Section 3 gave probabilities to events. But events are hard to compute with. What we really want is a *number* attached to a random outcome: a return, a count, a waiting time. Then we can average those numbers, measure their spread, and combine them. That number is a *random variable*. It is the most important object in the course. Every statistic of Section 7 is a random variable. All of inference studies how such numbers behave.

This section is long because it connects Section 3 to the rest of the course. It has seven parts.

- §4.1 defines random variables and their distributions.
- §4.2 defines the two summaries that matter most, expectation and variance.
- §4.3 defines the moments, the skewness and the kurtosis. It then introduces the moment generating function, which packages all the moments in one function.
- §4.4 gives two inequalities that bound probabilities from the mean and variance alone. The second is Chebyshev's, which Section 2 stated without proof. We prove it here.
- §4.5 covers two random variables together: their joint, marginal and conditional distributions.
- §4.6 is the most important part of this section. It covers expectations of two variables and independence. It then covers the law of iterated expectations, covariance and correlation. It ends with the mean and variance of a portfolio.
- §4.7 shows how a distribution changes when we apply a function to the variable.

Almost everything here is proved. A reader working alone should be able to reconstruct each argument.

### 4.1 Random Variables and Distributions

#### **Definition 4.1** (Random variable).

A random variable (隨機變數)  $X$  is a function  $X : S \rightarrow \mathbb{R}$  that assigns a real number to each outcome of a random experiment.

A random variable turns outcomes into numbers. So “the event that  $X$  lies in a set  $B$ ” means the event  $\{s \in S : X(s) \in B\}$ . The next example puts three random variables on one sample space.

**Remark (capital and small letters).** A capital letter,  $X$ , is the random variable: a function on  $S$ . A small letter,  $x$ , is one fixed number that  $X$  might take. So “ $X = x$ ” is an event, and  $x$  is not random.

**Example 4.2.** Toss a fair coin twice, so  $S = \{HH, HT, TH, TT\}$ . Define three variables. Let  $X$  be the number of heads. Let  $Y = 1$  if the first toss is a head and  $Y = 0$  otherwise. Let  $Z$  be the payoff of a bet that wins \$3 on  $HH$  and loses \$1 otherwise. (a) Tabulate  $X(s)$ ,  $Y(s)$  and  $Z(s)$  for every outcome  $s$ . (b) Write the event  $\{X = 1\}$  as a subset of  $S$  and find its probability. (c) Find the probability that  $Z = -1$ .

Outside the classroom the sample space is often too large to list. That does not matter. A stock's return tomorrow is a random variable. The outcome is tomorrow's state of the market.  $X$  is the return it produces. The number of defaults among ten loans is a random variable with values  $0, 1, \dots, 10$ . The waiting time for the next customer is a random variable with values in  $[0, \infty)$ . In each case there is a random experiment and a rule that gives a number.

**Proposition 4.3** (A function of a random variable).

Let  $X$  be a random variable and let  $g : \mathbb{R} \rightarrow \mathbb{R}$  be a function. Then  $Y = g(X)$  is also a random variable.

*Proof omitted.* The rule  $Y(s) = g(X(s))$  gives one real number for each outcome  $s$ . That is all Definition 4.1 asks for.

So  $X^2$ ,  $2X + 3$  and  $e^X$  are all random variables. For example, take the  $X$  of Example 4.2 and  $g(x) = x^2$ . Then  $X^2$  takes the values 4, 1, 1, 0 on the outcomes  $HH, HT, TH, TT$ . We use this fact often. §4.2 uses it to define the variance, and §4.7 studies  $g(X)$  in general.

The variable's *distribution* says how the probability is spread over the possible values. There are two basic kinds.

**Definition 4.4** (Discrete random variable; pmf).

$X$  is discrete if its possible values form a finite or countable set. Its probability mass function (機率質量函數) (pmf) is

$$p(x) = \quad , \quad \text{with } p(x) \geq 0 \text{ and } \sum_x p(x) = 1.$$

**Example 4.5.** Toss a fair coin three times and let  $X$  be the number of heads. Find the pmf of  $X$ .

**Example 4.6.** (a) Find the value of  $c$  that makes  $p_X$  a pmf:

$$p_X(x) = c(2x - 1), \quad x = 1, 2, \dots, 10.$$

(b) Let  $a > 0$  be a constant. Show that  $p_Y$  has the properties of a pmf:

$$p_Y(y) = \frac{e^{-a} a^y}{y!}, \quad y = 0, 1, 2, \dots$$

The probability mass function is how a discrete distribution is written down. Other distributions are spread out over an interval. We describe those by a density instead. For such a variable no single value carries positive probability.

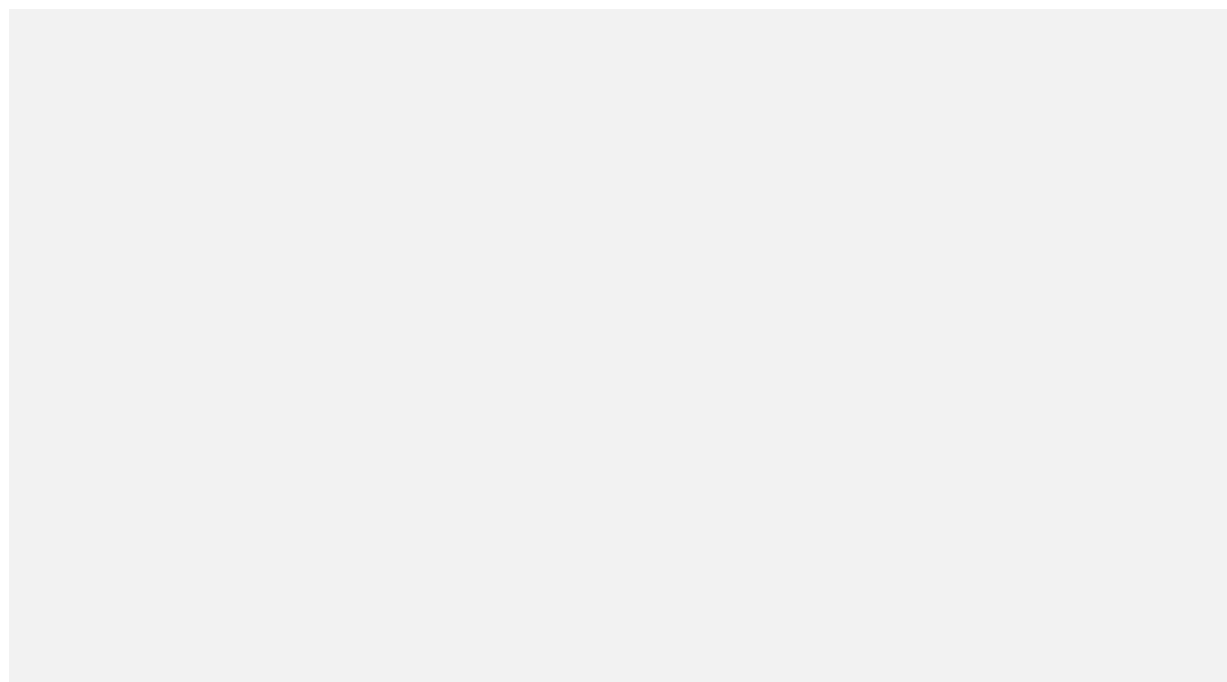
**Definition 4.7** (Continuous random variable; pdf).

$X$  is continuous if there is a function  $f \geq 0$ , its probability density function (機率密度函數) (pdf), with

$$P(a \leq X \leq b) = \int_a^b f(x) dx, \quad \int_{-\infty}^{\infty} f(x) dx = 1.$$

The first equation holds for every pair  $a \leq b$ . For a continuous  $X$  the probability of any single value is zero, so  $P(a \leq X \leq b) = P(a < X < b)$ .

**Example 4.8.** (a) Let  $f(x) = 2x$  for  $0 \leq x \leq 1$  and 0 otherwise. Verify that  $f$  is a density, and compute  $P(\frac{1}{2} < X < 1)$ . (b) Let  $f(x) = c x(1 - x)$  for  $0 \leq x \leq 1$  and 0 otherwise. Find the value of  $c$  that makes  $f$  a density. Then compute  $P(X > \frac{3}{4})$ , and find the height of  $f$  at  $x = \frac{1}{2}$ .



The probability density function is not itself a probability. It can exceed 1. Height at a point tells us nothing on its own. Figure 1 contrasts the two kinds. A discrete variable puts its probability on isolated points. A continuous variable spreads it along a curve.

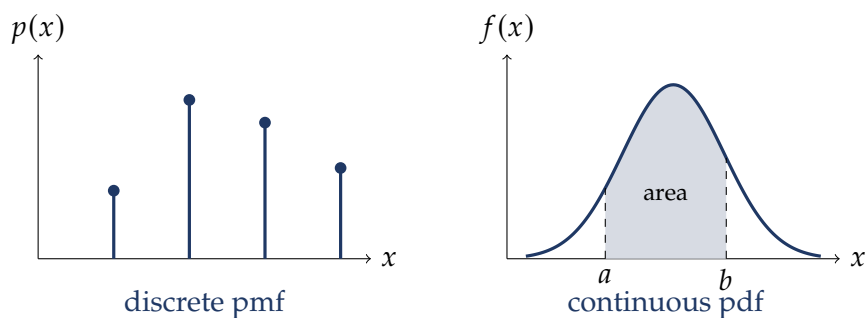


Figure 1: Two kinds of distribution. *Left:* a discrete pmf places probability at isolated points, drawn here as spikes. *Right:* a continuous pdf spreads probability along a curve, and the shaded region marks one interval.

Both kinds share one summary, the distribution function.

**Definition 4.9** (Cumulative distribution function).

The cumulative distribution function (累積分配函數) (CDF) of any random variable is

$$F(x) =$$

It is nondecreasing and right-continuous, with  $F(-\infty) = 0$  and  $F(+\infty) = 1$ . For continuous  $X$ ,  $F'(x) = f(x)$  wherever  $f$  is continuous.

The CDF is a step function for a discrete variable and a smooth curve for a continuous one (Figure 2). A common third case is a *mixed* (混合型) variable: part jump, part density. The remark and Example 4.15 below show one. First, one rule turns values of  $F$  into probabilities.

**Proposition 4.10** (Probabilities from the CDF).

For any random variable and any  $a < b$ ,

$$P(a < X \leq b) =$$

For a discrete variable with finitely many values, write  $F(x^-)$  for the value of  $F$  just to the left of  $x$ . Then, at each value  $x$ ,  $p(x) =$

*Proof omitted.* Both formulas come from the addition rule of Section 3. Figure 2 shows the first formula for a discrete variable and for a continuous variable. Its left panel also shows the second formula.

So a table of  $F$  answers every interval question. Sections 5 and 6 give such tables for the named distributions.

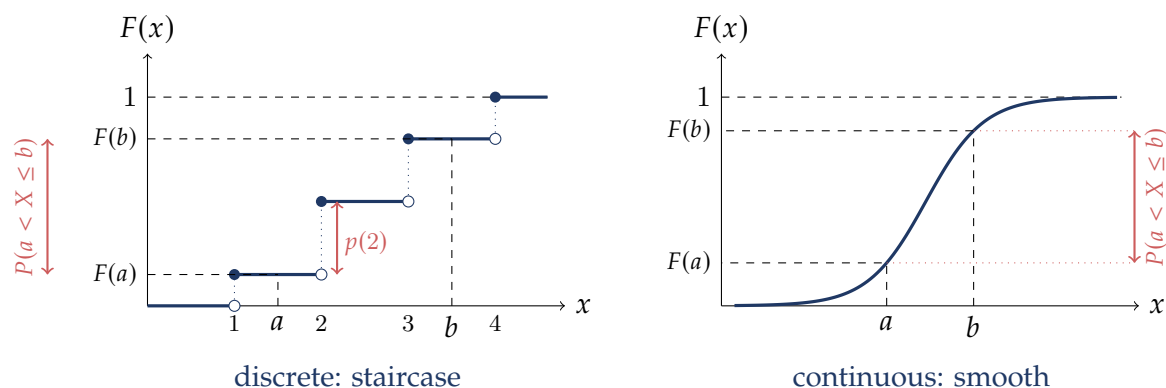


Figure 2: Two distribution functions, and Proposition 4.10 on each. *Left:* for a discrete variable  $F$  is a staircase. It is flat between values and jumps by  $p(x)$  at each value. The filled dots mark that  $F$  takes the upper value at a jump, which is right-continuity. *Right:* for a continuous variable  $F$  climbs smoothly from 0 to 1, with slope  $F'(x) = f(x)$ . In both panels the tall red arrow is the probability that  $X$  falls in  $(a, b]$ : the rise of  $F$  from  $a$  to  $b$ . The short red arrow in the left panel is the jump at  $x = 2$ .

**Example 4.11.** (a) For  $X =$  the number of heads in three tosses (Example 4.5), find the CDF and use it to compute  $P(X \geq 2)$ . (b) For the density  $f(x) = 2x$  on  $[0, 1]$  (Example 4.8), find the CDF and use it to compute  $P(X \leq \frac{1}{2})$  and  $P(\frac{1}{2} < X < 1)$ .

**Example 4.12.**  $X$  takes the values 1, 2, 3, 4 with  $p = (0.1, 0.2, 0.3, 0.4)$ . Find  $P(X < 2)$ ,  $P(X \leq 2)$ ,  $P(X \geq 2)$  and  $P(1 \leq X < 3)$ . Which of the four equals  $F(2)$ ?

**Example 4.13.** A discrete variable has  $F(x) = 0$  for  $x < 1$ ; 0.15 on  $[1, 2)$ ; 0.50 on  $[2, 3)$ ; 0.80 on  $[3, 4)$ ; and 1 for  $x \geq 4$ . This is the left panel of Figure 2. Recover the pmf, and find  $P(X > 2)$ .

**Example 4.14.** Let  $f(x) = x$  for  $0 \leq x \leq 1$ ,  $f(x) = 2 - x$  for  $1 < x \leq 2$ , and  $f(x) = 0$  otherwise. (a) Verify that  $f$  is a density. (b) Find  $F(x)$  on each piece. (c) Compute  $P(0.5 < X < 1.5)$ .

**Remark (mixed variables).** Take an insurance payout that is 0 with probability 0.7, and otherwise spread over  $(0, 1)$ . Its CDF jumps by 0.7 at 0, which is a point mass, and then rises continuously. We handle such a variable by combining a sum for the jumps with an integral for the density. Example 4.15 does exactly that.

**Example 4.15.** A payout  $X$  has the distribution function

$$F(x) = \begin{cases} 0, & x < 0, \\ 0.2, & 0 \leq x < 1, \\ 0.2 + 0.3(x - 1), & 1 \leq x < 2, \\ 0.7, & 2 \leq x < 3, \\ 0.7 + 0.2(x - 3), & 3 \leq x < 4, \\ 1, & x \geq 4. \end{cases}$$

(a) Sketch  $F$ . (b) Find  $P(X = 0)$ ,  $P(X = 2)$  and  $P(X = 4)$ . (c) Find the density of  $X$  on  $(1, 2)$  and on  $(3, 4)$ . (d) Find  $P(X \leq 1.5)$  and  $P(1.5 < X \leq 3.5)$ . (e) Find  $P(2 \leq X < 4)$ . (f) Check that the point masses and the density together carry probability 1.

## 4.2 Expectation and Variance

Throughout Section 4, every expectation, variance and covariance that appears is assumed finite. Every variance or standard deviation in a denominator is also assumed positive.

**Definition 4.16** (Expectation).

The expectation (期望值) (mean) of  $X$  is

$$\mathbb{E}(X) = \quad \quad \quad (\text{discrete}), \quad \quad \mathbb{E}(X) = \quad \quad \quad (\text{continuous}).$$

*This is the value that the average of many draws settles down to. It exists provided the sum or integral converges absolutely.*

**Example 4.17.** (a) Roll a fair die and let  $X$  be the face shown. Find  $\mathbb{E}(X)$ . (b) Let  $X$  have the pmf  $p_X(x) = \frac{2x-1}{100}$  for  $x = 1, 2, \dots, 10$ , as in Example 4.6. Find  $\mathbb{E}(X)$ .

**Example 4.18.** (a) Let  $f(x) = 2x$  for  $0 \leq x \leq 1$  and 0 otherwise (Example 4.8). Find  $\mathbb{E}(X)$ . (b) Let  $f(x) = 3/x^4$  for  $x \geq 1$  and 0 otherwise. Verify that  $f$  is a density and find  $\mathbb{E}(X)$ .

**Example 4.19.** A fair coin is tossed until the first head appears. If the first head appears on toss  $k$ , the game pays  $X = 2^k$  dollars. (a) Find the pmf of  $X$  and check that it sums to 1. (b) Find  $\mathbb{E}(X)$ . (c) Would you pay \$20 to play this game? Compare your answer with (b).

**Properties of the expectation.** We often need the mean of a function of  $X$ , such as  $X^2$  or  $e^X$ . By Proposition 4.3,  $g(X)$  is a random variable, so it has a mean of its own. The next result says how to find that mean. We do not need the distribution of  $g(X)$  first.

**Proposition 4.20** (Expectation of a function of  $X$ ).

For any function  $g$ ,

$$\mathbb{E}[g(X)] = \sum_{x \in \mathcal{X}} g(x) p_X(x) \quad (\text{discrete}), \quad \mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx \quad (\text{continuous}).$$

*Proof omitted.* In the discrete case the proof regroups the sum over the values of  $g(X)$ . The continuous case needs tools beyond this course. We use the result without proof.

This rule is often called the “law of the unconscious statistician”.

**Example 4.21.** A person values  $x$  dollars by the function  $u(x) = \sqrt{x}$ . The number  $u(x)$  is called the *utility* of  $x$  dollars. A gamble pays  $X = 100$  or  $X = 0$  dollars, each with probability  $\frac{1}{2}$ . (a) Find  $\mathbb{E}(X)$ . (b) Find the expected utility  $\mathbb{E}[u(X)]$ . (c) Compare  $\mathbb{E}[u(X)]$  with  $u(\mathbb{E}(X))$ . (d) Find the sure amount  $c$  with  $u(c) = \mathbb{E}[u(X)]$ . Would this person give up the gamble for a sure \$30?

**Example 4.22.** Return to the game of Example 4.19, where  $X = 2^k$  with probability  $(\frac{1}{2})^k$  for  $k = 1, 2, 3, \dots$ . A player values  $x$  dollars by  $u(x) = \ln x$ . (a) Show that  $\sum_{k=1}^{\infty} k(\frac{1}{2})^k = 2$ . (b) Find the expected utility  $\mathbb{E}[u(X)]$ . (c) Find the sure amount  $c$  with  $u(c) = \mathbb{E}[u(X)]$ , and compare with Example 4.19.

**Proposition 4.23** (Expectation of a linear transformation).

For constants  $a$  and  $b$ ,

$$\mathbb{E}(aX + b) = a \mathbb{E}(X) + b.$$

More generally, for functions  $g$  and  $h$ ,  $\mathbb{E}[a g(X) + b h(X)] = a \mathbb{E}[g(X)] + b \mathbb{E}[h(X)]$ .

*Proof.*

□

A change of units changes the mean in the same way. A sum can be taken apart term by term.

**Example 4.24.** (a) A game costs \$10 to enter and pays  $4X$  dollars, where  $X$  is the face of a fair die. Find the mean of the profit  $Y = 4X - 10$ . (b) Let  $X$  have the density  $f(x) = 3/x^4$  on  $[1, \infty)$  of Example 4.18(b). Find  $\mathbb{E}(X^2)$ , and then  $\mathbb{E}(2X^2 - 4X + 1)$ .

**Remark** ( $\mathbb{E}[g(X)]$  is **not**  $g(\mathbb{E}(X))$ ). Proposition 4.23 is special to linear functions. For other functions the two numbers differ in general:

$$\mathbb{E}(X^2) \neq (\mathbb{E}(X))^2, \quad \mathbb{E}\left(\frac{1}{X}\right) \neq \frac{1}{\mathbb{E}(X)}, \quad \mathbb{E}(e^X) \neq e^{\mathbb{E}(X)}.$$

Example 4.21(c) already showed this for the square root. Always compute  $\mathbb{E}[g(X)]$  from Proposition 4.20.

**Example 4.25.**  $X$  takes the values 1, 2, 4 with probabilities  $\frac{1}{2}, \frac{1}{4}, \frac{1}{4}$ . Compare  $\mathbb{E}(X^2)$  with  $(\mathbb{E}(X))^2$ ,  $\mathbb{E}(1/X)$  with  $1/\mathbb{E}(X)$ , and  $\mathbb{E}(e^X)$  with  $e^{\mathbb{E}(X)}$ .

**The variance and its properties.** Spread about the mean is measured by the variance.

**Definition 4.26** (Variance and standard deviation).

With  $\mu = \mathbb{E}(X)$ , the variance is

$$\text{Var}(X) =$$

The standard deviation is  $\sigma_X = \sqrt{\text{Var}(X)}$ .

Here the variance is itself the expectation of a function of  $X$ , so Proposition 4.20 computes it. The variance is the natural measure of spread about  $\mu$ . We almost always compute it from the following shortcut.

**Proposition 4.27** (Computing formula for the variance).

$\text{Var}(X) =$

*Proof.*

□

**Example 4.28.** For the die of Example 4.17, compute  $\text{Var}(X)$  twice: once from Definition 4.26, and once from Proposition 4.27.

**Proposition 4.29** (Mean and variance of an indicator).

Let  $A$  be an event with  $P(A) = p$ . Its indicator variable (指示變數)  $\mathbf{1}_A$  equals 1 if  $A$  occurs and 0 otherwise. Then  $\mathbb{E}(\mathbf{1}_A) = p$  and  $\text{Var}(\mathbf{1}_A) =$

*Proof.*

□

So every event gives a random variable. Every probability is also an expectation. For example,  $Y$  in Example 4.2 is the indicator of the event “the first toss is a head”. The variance is largest when the event is hardest to predict, at  $p = \frac{1}{2}$ . Section 5 calls this variable a Bernoulli variable.

**Proposition 4.30** (Variance of a linear transformation).

For constants  $a$  and  $b$ ,

$$\text{Var}(aX + b) =$$

Read this in words. Shifting by  $b$  slides the whole distribution sideways. It leaves the spread untouched. Scaling by  $a$  stretches every deviation from the mean. Spread therefore reacts to a scale change but not to a shift.

*Proof.*

□

**Corollary 4.31** (Mean zero, variance one).

Let  $X$  have mean  $\mu$  and standard deviation  $\sigma > 0$ , and set

$$Z = \frac{X - \mu}{\sigma}.$$

Then  $\mathbb{E}(Z) = 0$  and  $\text{Var}(Z) = 1$ .

*Proof.*

□

$Z$  counts how many standard deviations  $X$  sits from its mean. It is the population version of the z-score of Section 2. Section 6 applies this step to the normal distribution and gives it a name.

**Example 4.32.** Let  $X$  have the density  $f(x) = 6x(1 - x)$  on  $[0, 1]$  of Example 4.8(b). (a) Find  $\mathbb{E}(X)$  and  $\text{Var}(X)$ . (b) Find  $P(\frac{1}{4} < X < \frac{3}{4})$ . (c) Let  $Y = 10X - 2$ . Find  $\mathbb{E}(Y)$  and  $\text{Var}(Y)$ .

**Example 4.33.**  $X$  takes the values 0, 1, 2. It is known that  $\mathbb{E}(X) = 1.1$  and  $\mathbb{E}(X^2) = 1.7$ . Find the pmf of  $X$  and  $\text{Var}(X)$ .

### 4.3 Moments and the Moment Generating Function

The mean and variance are the first two of an infinite family of summaries. Every one of them is the expectation of a function of  $X$ . So Proposition 4.20 computes every one of them.

**Definition 4.34** (Raw and central moments).

Let  $k = 1, 2, 3, \dots$  and  $\mu = \mathbb{E}(X)$ . The  $k$ th raw moment (原動差) of  $X$  is  $\mu'_k = \mathbb{E}(X^k)$ . The  $k$ th central moment (主動差) of  $X$  is  $\mu_k = \mathbb{E}[(X - \mu)^k]$ .

A raw moment measures from 0. A central moment measures from the mean. The first raw moment is the mean:  $\mu'_1 = \mu$ . The first central moment is always 0, because  $\mu_1 = \mathbb{E}(X) - \mu = 0$ . The second central moment is the variance:  $\mu_2 = \sigma^2$ . Raw moments are easier to compute. The next result turns raw moments into central moments.

**Proposition 4.35** (Central moments from raw moments).

With  $\mu = \mu'_1$ ,

$$\mu_2 = \mu'_2 - \mu^2,$$

$$\mu_3 = \mu'_3 - 3\mu \mu'_2 + 2\mu^3,$$

$$\mu_4 = \mu'_4 - 4\mu \mu'_3 + 6\mu^2 \mu'_2 - 3\mu^4.$$

*Proof.*

□

The next definition gives two measures of the shape of a distribution.

**Definition 4.36** (Skewness and kurtosis).

Suppose  $0 < \sigma^2 < \infty$  and  $\mathbb{E}|X|^4 < \infty$ . The skewness  $\gamma_1$  and the kurtosis  $\gamma_2$  of  $X$  are

$$\gamma_1 = \frac{\mu_3}{\sigma^3}, \quad \gamma_2 = \frac{\mu_4}{\sigma^4}.$$

The excess kurtosis is  $\gamma_2 - 3$ .

**Reading the skewness.** Skewness measures asymmetry. This is the formula Section 2 promised. The formula keeps the sign of a deviation. It also makes a large deviation count much more than a small one.

- $\gamma_1 > 0$ : the large deviations above the mean count for more than the ones below. The right tail is usually longer. We say the distribution is skewed to the right.
- $\gamma_1 = 0$ : the two sides balance. Every symmetric distribution has  $\gamma_1 = 0$ .
- $\gamma_1 < 0$ : the large deviations below the mean count for more than the ones above. The left tail is usually longer. We say the distribution is skewed to the left.

Figure 3 shows the three cases.

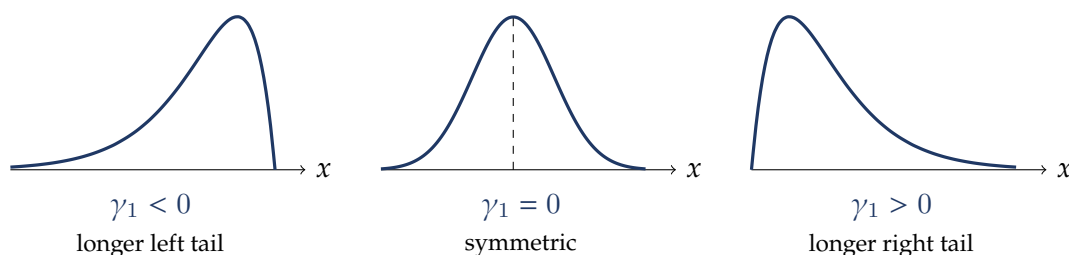


Figure 3: The sign of the skewness. *Left*: the long tail is on the left, so the large deviations are negative and  $\gamma_1 < 0$ . *Middle*: the two sides mirror each other, so  $\gamma_1 = 0$ . *Right*: the long tail is on the right, so  $\gamma_1 > 0$ .

**Reading the kurtosis.** Kurtosis measures how heavy the tails are. The formula gives a very large weight to a value far from the mean. So values far out in the tails decide the size of  $\gamma_2$ . We compare  $\gamma_2$  with 3. That is the kurtosis of the normal distribution, which Section 6 introduces.

- $\gamma_2 > 3$ : the tails are heavier than normal tails. Values far from the mean are more likely.
- $\gamma_2 = 3$ : the tails are as heavy as normal tails.
- $\gamma_2 < 3$ : the tails are lighter than normal tails. Values far from the mean are less likely.

These readings of  $\gamma_1$  and  $\gamma_2$  are the usual ones. They are guides, not theorems. Neither  $\gamma_1$  nor  $\gamma_2$  has units. So a change of units does not change them. Figure 4 shows three densities with the same mean and the same variance. In the figure, the density with  $\gamma_2 > 3$  also has the highest peak. But the kurtosis measures the tails, not the peak.

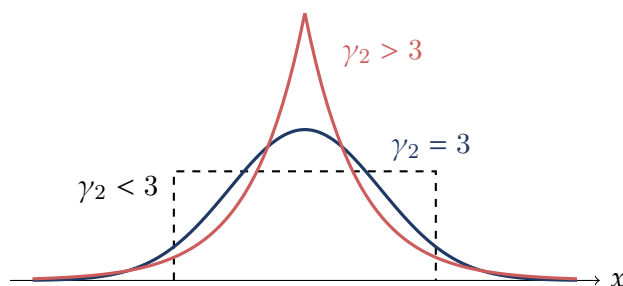


Figure 4: The size of the kurtosis. All three densities have the same mean and the same variance. The dashed density has no tails at all, and  $\gamma_2 < 3$ . The blue density is the normal one, with  $\gamma_2 = 3$ . The red density puts more probability far from the mean, and  $\gamma_2 > 3$ . At three standard deviations from the mean the red density is more than twice the blue one (0.0102 against 0.0044). The picture is too small to show that gap.

**Example 4.37.** A return  $X$  takes the values  $-1, 0, 2$  with probabilities  $0.2, 0.5, 0.3$ . (a) Find the raw moments  $\mu'_1, \mu'_2, \mu'_3, \mu'_4$ . (b) Use Proposition 4.35 to find  $\mu_2, \mu_3$  and  $\mu_4$ . (c) Find the skewness and the kurtosis, and say what each one tells us.

Computing moments one at a time is slow. We now build a single function that generates them all.

**Definition 4.38** (Moment generating function).

The moment generating function (動差生成函數) (MGF) of  $X$  is

$$M_X(t) =$$

This is defined for those  $t$  near 0 where the expectation is finite.

The MGF need not be finite for every  $t$ . We say the MGF *exists* if it is finite on an interval around 0. The next example shows an MGF that is finite for every  $t$ , and one that is not.

**Example 4.39.** (a) Let  $X$  be 1 with probability  $p$  and 0 with probability  $q = 1 - p$ . Find  $M_X(t)$ .  
(b) Let  $X$  have the density  $f(x) = e^{-x}$  for  $x \geq 0$  and 0 otherwise. Find  $M_X(t)$ , and say for which  $t$  it is finite.

The next result explains the name. It is why the MGF is worth defining at all.

**Proposition 4.40** (Where the name comes from).

If  $M_X$  is finite near 0, then  $M_X^{(k)}(0) =$

*Proof.*

□

**Example 4.41.** (a) Use the MGF of Example 4.39(a) to recover  $\mathbb{E}(X)$  and  $\text{Var}(X)$ , and compare with Proposition 4.29. (b) Let  $X$  have the pmf  $p(x) = \left(\frac{1}{2}\right)^x$  for  $x = 1, 2, 3, \dots$ . Find  $M_X(t)$ , and use it to find  $\mathbb{E}(X)$  and  $\text{Var}(X)$ .

**Theorem 4.42 (Uniqueness).**

*If  $X$  and  $Y$  have the same MGF on an interval around 0, they have the same distribution.*

*Proof omitted.* Recovering the distribution from  $M_X$  needs an inversion theorem for the Laplace or Fourier transform. That is beyond this course. We use uniqueness only to *identify* a distribution from its MGF, as in Example 4.44.

**Proposition 4.43** (MGF of a linear transformation).

For constants  $a, b$ ,  $M_{aX+b}(t) =$

*Proof.*

□

Section 6 uses this to pass between a normal variable and the  $Z$  of Corollary 4.31. Example 4.44 uses it on a three-point variable.

**Example 4.44.** A variable has  $M_X(t) = 0.2 + 0.5e^t + 0.3e^{3t}$ . (a) Read off the distribution of  $X$ . (b) Find  $\mathbb{E}(X)$  and  $\text{Var}(X)$  from the MGF. (c) Let  $Y = 2X + 1$ . Find  $M_Y(t)$ , and from it  $\mathbb{E}(Y)$  and  $\text{Var}(Y)$ .

#### 4.4 The Markov and Chebyshev Inequalities

How much probability can sit far from the mean? Two inequalities answer that. They use only the mean, or only the mean and the variance, whatever the distribution is.

**Theorem 4.45** (Markov's inequality).

*Markov's inequality* (馬可夫不等式) states that if  $X \geq 0$  and  $a > 0$ , then  $P(X \geq a) \leq$

*Proof.*

□

**Theorem 4.46** (Chebyshev's inequality).

*Chebyshev's inequality* (柴比雪夫不等式) states that for any  $X$  with mean  $\mu$  and finite variance  $\sigma^2 > 0$ , and any  $k > 0$ ,

$$P(|X - \mu| \geq k\sigma) \leq$$

*Proof.*

□

So Markov's inequality bounds a nonnegative variable by its mean. Applying it to  $(X - \mu)^2$  gives Chebyshev's, which uses the variance. Neither needs an assumption about the shape. So both are loose. Their value is that they hold for *every* distribution.

**Example 4.47.** Let  $X$  be the number of heads in three fair tosses, as in Example 4.5, so  $\mathbb{E}(X) = \frac{3}{2}$  and  $\text{Var}(X) = \frac{3}{4}$ . (a) Give Markov's bound on  $P(X \geq 3)$ , and the exact value. (b) Give Chebyshev's bound on  $P(|X - \mu| \geq 2\sigma)$ , and the exact value. (c) Say why the bounds are so much larger than the truth.

**Example 4.48.** Exam scores have mean 72 and standard deviation 4. (a) We want Chebyshev's inequality to guarantee that at least 90% of scores lie within  $k$  standard deviations of the mean. How large must  $k$  be? (b) Write down that interval.

**Example 4.49.** Let  $X$  take the values  $-1, 0, 1$  with probabilities  $\frac{1}{8}, \frac{3}{4}, \frac{1}{8}$ . (a) Find  $\mu$  and  $\sigma$ . (b) Compute  $P(|X - \mu| \geq 2\sigma)$  exactly and compare it with Chebyshev's bound. (c) What does the comparison say about improving the inequality?

## 4.5 Joint, Marginal and Conditional Distributions

Almost everything so far concerned one variable. But finance and inference both depend on how *two* variables move together.

**Definition 4.50** (Joint and marginal distributions).

For discrete  $X, Y$ , the joint pmf is  $p(x, y) = P(X = x, Y = y)$ . The marginal pmf of  $X$  is

$$p_X(x) =$$

For continuous  $X, Y$ , a joint pdf  $f(x, y) \geq 0$  plays the same role. The probability of a region  $A$  is the double integral of  $f$  over  $A$ . The integral over the whole plane is 1. The marginal pdf of  $X$  is

$$f_X(x) =$$

The *joint distribution* (聯合分配) describes the pair at once. The *marginal distribution* (邊際分配) describes one variable on its own. The marginals of  $Y$  are defined in the same way, with the roles of  $x$  and  $y$  exchanged.

Section 3 defined the conditional probability of an event  $A$  given an event  $B$ . The same idea works for two random variables. We fix the value of one variable and ask how the other one is distributed.

**Definition 4.51** (Conditional distribution, expectation and variance).

For  $p_X(x) > 0$ , the conditional pmf of  $Y$  given  $X = x$  is

$$p_{Y|X}(y | x) =$$

For continuous  $X, Y$  and  $f_X(x) > 0$ , the conditional pdf of  $Y$  given  $X = x$  is

$$f_{Y|X}(y | x) =$$

The conditional expectation (條件期望值)  $\mathbb{E}(Y | X = x)$  is the mean of this distribution:

$$\mathbb{E}(Y | X = x) = \sum_y y p_{Y|X}(y | x) \quad \text{or} \quad \int_{-\infty}^{\infty} y f_{Y|X}(y | x) dy.$$

The conditional variance is the variance of this distribution:  $\text{Var}(Y | X = x) = \mathbb{E}(Y^2 | X = x) - [\mathbb{E}(Y | X = x)]^2$ .

The *conditional distribution* (條件分配) of  $Y$  given  $X = x$  is an ordinary distribution. So it has an ordinary mean and an ordinary variance. They describe  $Y$  in the sub-population where  $X = x$ . Each value of  $x$  gives its own conditional distribution.

For continuous  $X, Y$  the event  $\{X = x\}$  has probability 0. So the formula of Section 3 cannot be used directly. We define a probability given  $X = x$  by the conditional pdf:  $P(a < Y < b | X = x) = \int_a^b f_{Y|X}(y | x) dy$ .

**Example 4.52.** The joint pmf of  $(X, Y)$  is

| $p(x, y)$ | $Y=0$ | $Y=1$ | $Y=2$ |
|-----------|-------|-------|-------|
| $X=0$     | 0.10  | 0.08  | 0.02  |
| $X=1$     | 0.10  | 0.25  | 0.15  |
| $X=2$     | 0.03  | 0.09  | 0.18  |

(a) Find the marginal pmf of  $X$  and the marginal pmf of  $Y$ . (b) Find  $P(X = Y)$  and  $P(X < Y)$ . (c) Find the conditional pmf of  $Y$  given  $X = x$ , for  $x = 0, 1, 2$ . (d) Find  $\mathbb{E}(Y \mid X = x)$  and  $\text{Var}(Y \mid X = x)$  for each  $x$ . (e) Find  $P(X \geq 1 \mid Y = 2)$ . (f) Let  $S = X + Y$ . Find the pmf of  $S$  and  $\mathbb{E}(S)$ .

One function describes a pair of variables of any kind, as the CDF did for one variable.

**Definition 4.53** (Joint cumulative distribution function).

The joint cumulative distribution function (聯合累積分配函數) (joint CDF) of  $X$  and  $Y$  is

$$F(x, y) = P(X \leq x, Y \leq y).$$

For continuous  $X, Y$  with joint pdf  $f$ ,

$$F(x, y) = \int_{-\infty}^x \int_{-\infty}^y f(s, t) dt ds.$$

For a continuous pair, the joint pdf is a surface over the plane. A probability is a volume under that surface. Figure 5 shows a joint pdf and its joint CDF.

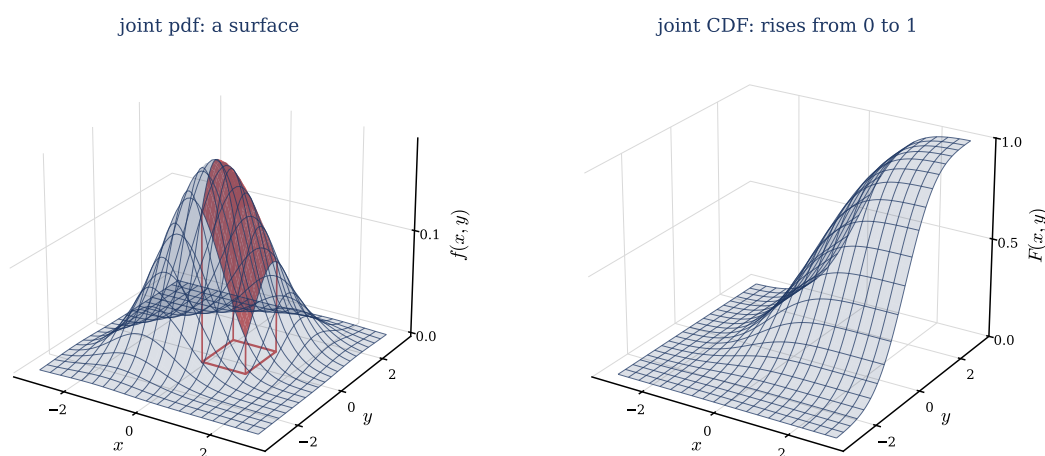


Figure 5: A joint pdf and its joint CDF. *Left*: the joint pdf is a surface. The probability of the rectangle on the floor is the volume of the red column above it. *Right*: the joint CDF of the same pair. It is near 0 when  $x$  and  $y$  are both small, and it rises to 1 when both are large.

$F(x, y)$  is the probability of the region to the left of  $x$  and below  $y$ . The left panel of Figure 6 shows that region. With one variable,  $F$  gave the probability of an interval. With two variables,  $F$  gives the probability of a rectangle.

**Proposition 4.54** (Probability of a rectangle).

For  $a < b$  and  $c < d$ ,

$$P(a < X \leq b, c < Y \leq d) = F(b, d) - F(a, d) - F(b, c) + F(a, c).$$

*Proof.*

□

The right panel of Figure 6 shows the four pieces. We subtract two strips. Their overlap was subtracted twice, so we add it back.

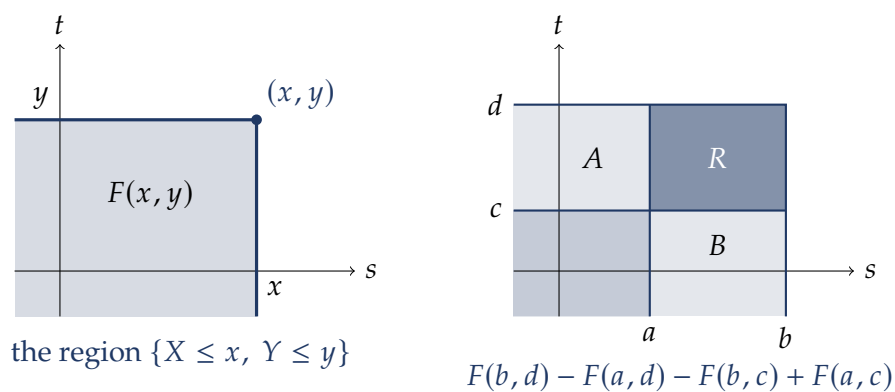


Figure 6: The joint CDF. *Left:*  $F(x, y)$  is the probability that the pair falls in the shaded region, to the left of  $x$  and below  $y$ . *Right:* start from the region for  $(b, d)$ . Remove the strip  $A$  and the strip  $B$ . The dark rectangle  $R$  is left. The lower-left corner is in both strips, so it is added back once.

**Example 4.55.**  $X$  and  $Y$  have the joint pdf  $f(x, y) = c(x + y)$  for  $0 < x < 2$  and  $0 < y < 1$ , and  $f(x, y) = 0$  elsewhere. (a) Find  $c$ . (b) Find the marginal pdf of  $X$  and the marginal pdf of  $Y$ . (c) Find the joint CDF  $F(x, y)$  for  $0 \leq x \leq 2$  and  $0 \leq y \leq 1$ . Use it to find  $P(1 < X \leq 2, \frac{1}{2} < Y \leq 1)$ . (d) Find the conditional pdf of  $X$  given  $Y = y$ , and then  $P(\frac{1}{2} < X < \frac{3}{2} \mid Y = \frac{1}{2})$ . (e) Find  $\mathbb{E}(X \mid Y = \frac{1}{2})$ . (f) Find  $P(X + Y > 1)$ .

## 4.6 Expectation, Independence, Covariance and Correlation

§4.2 gave the rules for the mean of one variable. The same rules hold for two variables. We weight a function of the pair by the joint distribution.

**Proposition 4.56** (Expectation of a function of two variables).

For any function  $g$  of two variables,

$$\mathbb{E}[g(X, Y)] = \sum_x \sum_y g(x, y) p(x, y) \quad (\text{discrete}),$$

$$\mathbb{E}[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) dy dx \quad (\text{continuous}).$$

*Proof omitted.* This is Proposition 4.20 for a pair of variables. We use it without proof.

**Theorem 4.57** (Linearity of expectation).

For constants  $a, b$ ,  $\mathbb{E}(aX + bY) = a \mathbb{E}(X) + b \mathbb{E}(Y)$ .

*Proof.*

□

The second summary of §4.2 was the variance. The variance of  $aX + bY$  needs one more idea, the covariance. It comes at the end of this subsection, in Theorem 4.73.

§4.3 put all the moments of one variable into one function. The same can be done for a pair.

**Definition 4.58** (Joint moment generating function).

The joint moment generating function (聯合動差生成函數) (joint MGF) of  $X$  and  $Y$  is

$$M_{X,Y}(s, t) = \mathbb{E}(e^{sX+tY}).$$

It is defined for those  $(s, t)$  near  $(0, 0)$  where the expectation is finite.

**Proposition 4.59** (What the joint MGF gives).

Let  $M_{X,Y}$  be finite near  $(0,0)$ . Then  $M_{X,Y}(s,0) = M_X(s)$  and  $M_{X,Y}(0,t) = M_Y(t)$ . Differentiate  $M_{X,Y}$  once in  $s$  and once in  $t$ , and then set  $s = t = 0$ . The result is  $\mathbb{E}(XY)$ .

*Proof.*

□

**Example 4.60.** The joint pmf of  $(X, Y)$  is  $p(0,0) = 0.3, p(0,1) = 0.2, p(1,0) = 0.1, p(1,1) = 0.4$ . (a) Find the joint MGF  $M_{X,Y}(s,t)$ . (b) Use it to find  $M_X(s)$  and  $M_Y(t)$ , and then  $\mathbb{E}(X)$  and  $\mathbb{E}(Y)$ . (c) Use it to find  $\mathbb{E}(XY)$ .

The next definition treats the most important special case of a joint distribution.

**Definition 4.61** (Independence of random variables).

$X$  and  $Y$  are independent if  $p(x, y) =$  \_\_\_\_\_ for all  $x, y$ . For continuous  $X, Y$  the condition is  $f(x, y) =$  \_\_\_\_\_

Suppose  $X$  and  $Y$  are independent. Then by Definition 4.51 the conditional pmf of  $Y$  given  $X = x$  is the marginal pmf of  $Y$ , for every  $x$  with  $p_X(x) > 0$ . So learning the value of  $X$  changes nothing about the distribution of  $Y$ . This is the strongest way to say that  $X$  and  $Y$  carry no information about each other. In Example 4.52 the conditional pmf of  $Y$  changes with  $x$ . So the  $X$  and  $Y$  of that example are not independent.

Independence has one more consequence, and many later proofs need it.

**Proposition 4.62** (Functions of independent variables).

If  $X$  and  $Y$  are independent, then for any functions  $g$  and  $h$ ,

$$\mathbb{E}[g(X) h(Y)] =$$

This covers  $g(x) = x$  and  $h(y) = y$  in particular.

*Proof.*

□

Independence also makes the MGF of a sum easy. §4.3 could not state this result, because it needs two variables.

**Proposition 4.63** (MGF of a sum of independent variables).

If  $X$  and  $Y$  are independent, then  $M_{X+Y}(t) =$

*Proof.*

□

Sections 5 to 7 all use this property. The distribution of a *sum* of independent variables follows from the two separate MGFs. That replaces a hard direct calculation with an easy one-step one. Uniqueness (Theorem 4.42) then reads off the answer.

**Example 4.64.** Let  $X_1, X_2$  be independent, each as in Example 4.39(a): 1 with probability  $p$  and 0 with probability  $q = 1 - p$ . Find the MGF of  $X_1 + X_2$  and say what distribution it identifies.

**Example 4.65.**  $X$  and  $Y$  are independent.  $X$  has the pdf  $f_X(x) = 2x$  on  $(0, 1)$ .  $Y$  has the pdf  $f_Y(y) = e^{-y}$  on  $(0, \infty)$ . (a) Write down the joint pdf. (b) Find  $P(X > \frac{1}{2}, Y > 1)$ . (c) Find  $\mathbb{E}(XY)$  and  $\mathbb{E}(X^2e^{-Y})$ . (d) Find  $P(Y > X)$ .

§4.5 defined  $\mathbb{E}(Y \mid X = x)$  for each value  $x$ . Now let  $x$  vary. Then  $\mathbb{E}(Y \mid X)$  is itself a random variable, because it is a function of  $X$ . In the same way,  $\text{Var}(Y \mid X)$  is the random variable that equals  $\text{Var}(Y \mid X = x)$  when  $X = x$ . Now average  $\mathbb{E}(Y \mid X)$  over the distribution of  $X$ . That gives back something we already know.

**Theorem 4.66** (Law of iterated expectations).

The law of iterated expectations (重複期望值定律) states that  $\mathbb{E}[\mathbb{E}(Y \mid X)] =$

*Proof.*

□

**Theorem 4.67** (Law of total variance).

The law of total variance (全變異數定律) states that

$$\text{Var}(Y) =$$

*Proof.*

□

Read the two pieces as a split of  $Y$ 's variability. One is the spread that survives inside each value of  $X$ . The other is the spread of the group means as  $X$  moves. Both are nonnegative. So on average, conditioning on  $X$  can only remove variability from  $Y$ , never add it. For one particular value of  $X$  the conditional variance can still be larger than  $\text{Var}(Y)$ .

**Proposition 4.68** (Law of total probability for random variables).

For discrete  $X, Y$ ,

$$p_Y(y) = \sum_x p_{Y|X}(y | x) p_X(x).$$

For continuous  $X, Y$ ,  $f_Y(y) = \int_{-\infty}^{\infty} f_{Y|X}(y | x) f_X(x) dx$ .

*Proof.*

□

This is the law of total probability of Section 3, written for random variables. The three laws do the same job. Each one rebuilds a fact about  $Y$  from the conditional distributions given  $X$ .

**Example 4.69.**  $X$  is equally likely to be 1, 2 or 3. Given  $X = x$ , the variable  $Y$  is equally likely to be any of the values  $1, \dots, x$ . (a) Find the marginal pmf of  $Y$  with the law of total probability. (b) Find  $\mathbb{E}(Y \mid X = x)$  for each  $x$ , and then  $\mathbb{E}(Y)$  with the law of iterated expectations. (c) Find  $\text{Var}(Y \mid X = x)$  for each  $x$ , and then  $\text{Var}(Y)$  with the law of total variance. (d) Check  $\mathbb{E}(Y)$  and  $\text{Var}(Y)$  from the pmf of  $Y$ .

We measure the strength and direction of a linear relationship exactly as in Section 2. The difference is that we now do it for the theoretical distribution.

**Definition 4.70** (Covariance and correlation).

With  $\mu_X = \mathbb{E}(X)$  and  $\mu_Y = \mathbb{E}(Y)$ ,

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)], \quad \rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} \in [-1, 1].$$

No positivity condition is needed for the covariance. It is defined whenever both variances are finite. But  $\rho_{XY}$  needs both standard deviations to be positive and finite.

The covariance gives the direction of co-movement. Its units are the units of  $X$  times the units of  $Y$ . The \_\_\_\_\_ rescales it to  $[-1, 1]$ , so it has no units. Proposition 4.74 below proves that bound. We almost always use the shortcut below.

**Proposition 4.71** (Covariance shortcut).

$\text{Cov}(X, Y) =$

*Proof.*

□

**Proposition 4.72** (Properties of the covariance).

For constants  $a, b, c, d$ ,

$$\text{Cov}(aX + b, cY + d) =$$

Also  $\text{Cov}(X, Y) = \text{Cov}(Y, X)$ ,  $\text{Cov}(X, X) = \text{Var}(X)$ , and  $\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$ .

If  $a$  and  $c$  have the same sign, then  $\rho_{aX+b, cY+d} = \rho_{XY}$ . If they have opposite signs,  $\rho_{aX+b, cY+d} = -\rho_{XY}$ .

*Proof.*

□

So a change of units, such as percent to decimal, changes the covariance but not  $\rho_{XY}$ . Section 2 claimed this for the sample version; here it is proved for the distribution.

The variance of a sum needs the covariance.

**Theorem 4.73** (Variance of a linear combination).

$$\text{Var}(aX + bY) =$$

Two variables that move together do not simply add their spreads. The sign of the covariance decides whether the combination is more variable or less. Suppose the coefficients share a sign, as portfolio weights do. Then positive covariance inflates the variance, and negative covariance cancels part of it. That is the mechanism behind the portfolio result below. With coefficients of opposite sign this reverses, as in Example 4.76(c).

*Proof.*

□

**Proposition 4.74** (The bound on  $\rho$ ).

$-1 \leq \rho_{XY} \leq 1$ . Moreover,  $\rho_{XY} = 1$  if and only if  $Y = aX + b$  with probability 1 for some constants  $a > 0$  and  $b$ . Likewise  $\rho_{XY} = -1$  if and only if the same holds with  $a < 0$ .

*Proof.*

□

So  $\rho_{XY}$  measures a *linear* relationship, and  $\pm 1$  means an exact line. Section 2 could only assert this for the sample coefficient.

**Proposition 4.75** (Independence gives zero covariance).

If  $X$  and  $Y$  are independent, then  $\text{Cov}(X, Y) = 0$ . The converse is false.

*Proof.*

□

**Example 4.76.** Return to the joint pmf of Example 4.52, with marginals  $p_X = (0.20, 0.50, 0.30)$  and  $p_Y = (0.23, 0.42, 0.35)$  on the values 0, 1, 2. (a) Find  $\mathbb{E}(X)$ ,  $\mathbb{E}(Y)$ ,  $\text{Var}(X)$  and  $\text{Var}(Y)$ . (b) Find  $\mathbb{E}(XY)$ ,  $\text{Cov}(X, Y)$  and  $\rho_{XY}$ . (c) Find  $\text{Var}(X + Y)$  and  $\text{Var}(2X - Y)$ . (d) Let  $U = 2X + 1$  and  $V = -3Y + 5$ . Find  $\text{Cov}(U, V)$  and  $\rho_{UV}$ . (e) Are  $X$  and  $Y$  independent?

A portfolio is the case that matters here. Put weights  $w_1, w_2$  on two assets with returns  $R_1, R_2$ . The portfolio return is  $R_p = w_1 R_1 + w_2 R_2$ . Its mean and variance come straight from Theorems 4.57 and 4.73. Take positive weights. Whenever  $\rho < 1$ , the portfolio's risk falls below the weighted average of the two risks. That is \_\_\_\_\_. Falling below the *smaller* of the two risks needs more: a small enough  $\rho$ , and the right weights. Example 4.77 shows both parts.

**Example 4.77.** Two assets have  $\mathbb{E}(R_1) = 8\%$ ,  $\mathbb{E}(R_2) = 12\%$ ,  $\sigma_1 = 10\%$ ,  $\sigma_2 = 20\%$  and  $\rho = 0.3$ . (a) A portfolio puts 60% in asset 1 and 40% in asset 2. Find its expected return and its standard deviation. Compare with the weighted average of the two standard deviations. (b) Now put weight  $w$  on asset 1 and  $1-w$  on asset 2. Write  $\text{Var}(R_p)$  as a function  $V(w)$ . (c) Find the weight  $w^*$  that minimizes  $V$ , and the resulting  $\sigma_p$ . Is it below the smaller risk, 10%? (d) Repeat (c) with  $\rho = -1$ .

The rest of the course works with many variables at once, not two. The next definition and two propositions extend what we have to  $n$  variables.

**Definition 4.78** (Independent and identically distributed variables).

Random variables  $X_1, \dots, X_n$  have the joint pmf  $p(x_1, \dots, x_n) = P(X_1 = x_1, \dots, X_n = x_n)$ , defined as for two variables. They are independent if this joint pmf (or joint pdf) factors into the product of the  $n$  marginals. This must hold for every choice of values. They are independent and identically distributed (獨立且相同分配) (i.i.d.) if, in addition, all  $n$  have the same distribution.

Sampling with replacement from a population gives i.i.d. draws. Each draw has the population's distribution, and no draw affects another. Section 7 builds every sampling distribution on such draws.

**Proposition 4.79** (Mean and variance of a sum).

For random variables  $X_1, \dots, X_n$  and constants  $a_1, \dots, a_n$ ,

$$\mathbb{E}\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i \mathbb{E}(X_i), \quad \text{Var}\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i^2 \text{Var}(X_i) + 2 \sum_{i < j} a_i a_j \text{Cov}(X_i, X_j).$$

If  $X_1, \dots, X_n$  are independent, every covariance term is zero.

*Proof.*

□

**Proposition 4.80** (Sums of i.i.d. variables).

Let  $X_1, \dots, X_n$  be i.i.d., each with mean  $\mu$ , variance  $\sigma^2$  and MGF  $M_X$ , and let  $S = X_1 + \dots + X_n$ . Then

$$\mathbb{E}(S) = n\mu, \quad \text{Var}(S) = \quad, \quad M_S(t) =$$

*Proof.*

□

Section 5 builds the binomial from  $n$  Bernoulli variables with the MGF part. Section 7 divides  $S$  by  $n$  and studies the sample mean.

**Example 4.81.** In one hour a shop serves  $N$  customers, where  $N$  takes the values 0, 1, 2 with probabilities 0.2, 0.5, 0.3. Each customer's spend has mean \$40 and standard deviation \$20. The spends are independent of each other and of  $N$ . Let  $T$  be the hour's total takings. (a) Find  $\mathbb{E}(T \mid N = n)$  and  $\text{Var}(T \mid N = n)$ . (b) Use the law of iterated expectations to find  $\mathbb{E}(T)$ . (c) Find  $\text{Var}(N)$ . (d) Use the law of total variance to find  $\text{Var}(T)$  and the standard deviation of  $T$ .

### 4.7 Transformations of Random Variables

Suppose we know the distribution of  $X$ . What is the distribution of  $Y = g(X)$ ? Applying a function to a random variable is called a change of variable (變數變換). There are four methods.

- **List the values.** For a discrete  $X$ , collect the mass of every  $x$  that maps to each  $y$ :  $p_Y(y) = \sum_{x: g(x)=y} p_X(x)$ .
- **The CDF method.** Write  $F_Y(y) = P(g(X) \leq y)$  as a probability about  $X$ . Then differentiate to get  $f_Y$ . The first step always works. The derivative is a pdf when  $Y$  is continuous.
- **The MGF method.** Find  $M_Y(t)$ . Then find a distribution with that MGF. By uniqueness (Theorem 4.42) it is the distribution of  $Y$ . This method is best for sums of independent variables, as in Example 4.64.
- **The Jacobian method.** A formula gives  $f_Y$  directly from  $f_X$  and the derivative of the inverse map. Proposition 4.83 is the case of one variable.

**Example 4.82.**  $X$  takes values  $-1, 0, 1, 2$  with probabilities  $0.1, 0.4, 0.3, 0.2$ . Find the pmf of  $Y = X^2$ .

**Proposition 4.83** (Monotone change of variable).

Let  $g$  be strictly monotone and differentiable, with  $g'(x) \neq 0$  wherever  $f_X(x) > 0$ , and with inverse  $x = g^{-1}(y)$ . Take any  $y = g(x)$  with  $f_X(x) > 0$ . Then

$$f_Y(y) =$$

For every other  $y$ ,  $f_Y(y) = 0$ .

*Proof.*

□

The proposition needs a monotone  $g$ . If  $g$  is not monotone, split the range of  $X$  into pieces on which  $g$  is monotone. Apply the proposition on each piece. Then add the results for each  $y$ . The pieces are disjoint events, so their contributions add.

**Example 4.84.**  $X$  has the pdf  $f_X(x) = \frac{1}{5}$  on  $-3 \leq x \leq 2$ , and 0 elsewhere. Let  $Y = X^2$ . (a) Find the pdf of  $Y$  with the CDF method. (b) Find it again with Proposition 4.83, used on two pieces. (c) Check that the pdf of  $Y$  integrates to 1.

**Example 4.85.**  $X$  has the pdf  $f_X(x) = e^{-x}$  for  $x > 0$ , and 0 elsewhere. Let  $Y = 2X$ . Find the pdf of  $Y$  in three ways: (a) with the CDF method, (b) with Proposition 4.83, (c) with the MGF method.

**Example 4.86.**  $X$  has the pdf  $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$  for all real  $x$ . Take as given that  $f_X$  is a pdf. Let  $Y = X^2$ . (a) Find the pdf of  $Y$  with the CDF method. (b) Find it again with Proposition 4.83, used on two pieces.

For *two* variables the same idea uses the \_\_\_\_\_. Under a smooth invertible map  $(X, Y) \rightarrow (U, V)$  we have  $f_{U,V}(u, v) = f_{X,Y}(x, y) |J|$ , where  $J = \det \frac{\partial(x, y)}{\partial(u, v)}$ . Here  $|J|$  converts area in the  $(u, v)$  plane into area in the  $(x, y)$  plane. We show this on two concrete examples rather than prove the general theorem. The work has four steps.

1. Solve for the old variables: write  $x$  and  $y$  in terms of  $u$  and  $v$ .
2. Compute  $J$  from the four partial derivatives.
3. Write  $f_{U,V}(u, v) = f_{X,Y}(x, y) |J|$ , with  $x$  and  $y$  replaced.
4. Find the region of  $(u, v)$  where this pdf is positive. A pdf is not complete without its region.

If the question asks for the pdf of  $U$  alone, integrate out  $v$  at the end.

**Example 4.87.**  $X$  and  $Y$  are independent, each with the pdf  $f(x) = 1$  on  $(0, 1)$ . So  $f_{X,Y}(x, y) = 1$  on the unit square. Let  $U = X + Y$  and  $V = X - Y$ . (a) Find the joint pdf of  $(U, V)$  and the region where it is positive. (b) Find the pdf of  $U$ . (c) Check (b) with the CDF method.

**Example 4.88.**  $X_1$  and  $X_2$  have the joint pdf  $f(x_1, x_2) = 4x_1x_2$  for  $0 < x_1 < 1$  and  $0 < x_2 < 1$ , and 0 elsewhere. Let  $Y_1 = X_1/X_2$  and  $Y_2 = X_1X_2$ . (a) Find the joint pdf of  $(Y_1, Y_2)$  and the region where it is positive. (b) Find the pdf of  $Y_1$ . (c) Find the pdf of  $Y_2$ . (d) Check the pdf of  $Y_1$  on  $0 < y_1 \leq 1$  with the CDF method.

## Exercises

**Exercise 4.1.** A discrete random variable  $X$  has  $p(1) = 0.2$ ,  $p(2) = 0.5$ ,  $p(3) = 0.3$ . Verify this is a valid pmf, find  $F(2)$ , and find  $P(X \geq 2)$ .

**Exercise 4.2.** A discrete random variable has  $p(x) = c(4 - x)$  for  $x = 0, 1, 2, 3$ . Find  $c$ ,  $\mathbb{E}(X)$ ,  $\text{Var}(X)$  and  $P(X \geq 2)$ .

**Exercise 4.3.** For the  $X$  of Exercise 4.1, find  $\mathbb{E}(X)$ ,  $\text{Var}(X)$ , and  $\mathbb{E}(5X - 2)$ .

**Exercise 4.4.** Let  $f(x) = cx^2$  for  $0 \leq x \leq 1$  and 0 otherwise. Find  $c$ , the CDF, and  $P(X \leq \frac{1}{2})$ .

**Exercise 4.5.** A continuous random variable has CDF  $F(x) = x^2/9$  for  $0 \leq x \leq 3$ , with  $F(x) = 0$  for  $x < 0$  and  $F(x) = 1$  for  $x > 3$ . Find the density,  $P(1 < X < 2)$ , the median (the value  $m$  with  $F(m) = \frac{1}{2}$ ), and  $\mathbb{E}(X)$ .

**Exercise 4.6.** For the  $X$  of Exercise 4.4, find  $\mathbb{E}(X)$  and  $\text{Var}(X)$ .

**Exercise 4.7.** Let  $f(x) = c(1 - x^2)$  for  $-1 \leq x \leq 1$  and 0 otherwise. Find  $c$ ,  $\mathbb{E}(X)$ ,  $\text{Var}(X)$  and  $P(X > \frac{1}{2})$ .

**Exercise 4.8.**  $X$  takes the values 1, 2, 3. It is known that  $\mathbb{E}(X) = 2.2$  and  $\text{Var}(X) = 0.56$ . Find the pmf of  $X$ .

**Exercise 4.9.** A random variable has MGF  $M_X(t) = 0.4 + 0.6e^t$ . Find  $\mathbb{E}(X)$  and  $\text{Var}(X)$ . Then say which two values  $X$  takes, and with what probabilities.

**Exercise 4.10.** A random variable has  $M_X(t) = \frac{1}{4}(1 + e^t)^2$ . (a) Read off the distribution of  $X$ , and find  $\mathbb{E}(X)$  and  $\text{Var}(X)$  from the MGF. (b) Let  $Y = 3X - 2$ . Find  $M_Y(t)$ ,  $\mathbb{E}(Y)$  and  $\text{Var}(Y)$ .

**Exercise 4.11.** (a) Let  $X$  be 1 with probability 0.2 and 0 with probability 0.8. Find the skewness  $\gamma_1$ . (b) Let  $X$  be  $-1$  or  $1$  with probability  $\frac{1}{2}$  each. Find the kurtosis  $\gamma_2$  and the excess kurtosis.

**Exercise 4.12.** A nonnegative random variable has mean 10 and standard deviation 2. Give the Markov bound on  $P(X \geq 25)$  and the Chebyshev bound on  $P(|X - 10| \geq 6)$ .

**Exercise 4.13.** A nonnegative random variable has mean 4. (a) Give Markov's bound on  $P(X \geq 10)$ . (b) Construct a distribution with mean 4 for which  $P(X \geq 10)$  equals that bound. What does this say about improving the inequality?

**Exercise 4.14.** Roll a fair die and let  $X$  be the face shown, so  $\mu = 3.5$  and  $\sigma^2 = \frac{35}{12}$  (Example 4.28). Let  $Z = (X - \mu)/\sigma$ . (a) State  $\mathbb{E}(Z)$  and  $\text{Var}(Z)$ . (b) Compute  $P(|Z| \geq 1.4)$  exactly and compare it with Chebyshev's bound.

**Exercise 4.15.** The joint pmf of  $(X, Y)$  is  $p(0, 0) = 0.4$ ,  $p(0, 1) = 0.1$ ,  $p(1, 0) = 0.1$ ,  $p(1, 1) = 0.4$ . Find the marginals,  $\text{Cov}(X, Y)$ , and  $\rho_{XY}$ , and state whether  $X, Y$  are independent.

**Exercise 4.16.** Using the joint pmf of Exercise 4.15, find  $\mathbb{E}(X + Y)$  and  $\text{Var}(X + Y)$ .

**Exercise 4.17.**  $X$  has marginal pmf  $p_X = (0.2, 0.5, 0.3)$  on  $\{0, 1, 2\}$ .  $Y$  has marginal pmf  $p_Y = (0.4, 0.6)$  on  $\{0, 1\}$ .  $X$  and  $Y$  are independent. (a) Write out the joint pmf as a table. (b) Verify  $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$  from the table.

**Exercise 4.18.** The joint pmf of  $(X, Y)$  is  $p(0, 0) = 0.1$ ,  $p(0, 1) = 0.3$ ,  $p(1, 0) = 0.4$ ,  $p(1, 1) = 0.2$ . Find  $\text{Cov}(X, Y)$ ,  $\rho_{XY}$ ,  $\text{Var}(X - Y)$ ,  $\text{Var}(X + 2Y)$ , and the correlation between  $2X + 1$  and  $-3Y + 5$ .

**Exercise 4.19.** Two assets have  $\mathbb{E}(R_1) = 6\%$ ,  $\mathbb{E}(R_2) = 10\%$ ,  $\sigma_1 = 8\%$ ,  $\sigma_2 = 15\%$ , and  $\rho = 0.5$ . An equal-weighted portfolio ( $w_1 = w_2 = 0.5$ ) is formed. Find  $\mathbb{E}(R_p)$ ,  $\text{Var}(R_p)$ , and  $\sigma_p$ , and compare  $\sigma_p$  with the weighted average of the two standard deviations.

**Exercise 4.20.** Three assets each have standard deviation 10%, and each pair has correlation  $\rho = 0.2$ . A portfolio puts weight  $\frac{1}{3}$  on each. (a) Find  $\sigma_p$ . (b) Repeat with  $\rho = 0$ . Compare both with the single-asset risk.

**Exercise 4.21.** Two assets have  $\sigma_1 = 6\%$ ,  $\sigma_2 = 12\%$  and  $\rho = 0.25$ . Find the weight  $w$  on asset 1 that minimizes the portfolio variance, and the resulting  $\sigma_p$ . Is it below 6%?

**Exercise 4.22.**  $X_1, \dots, X_{25}$  are i.i.d. with mean 2, variance 9 and MGF  $M_X$ . Let  $S = X_1 + \dots + X_{25}$ . Find  $\mathbb{E}(S)$ ,  $\text{Var}(S)$  and  $M_S(t)$ , and give Chebyshev's bound on  $P(|S - 50| \geq 30)$ .

**Exercise 4.23.** The joint pmf of  $(X, Y)$  is  $p(0, 0) = 0.2$ ,  $p(0, 1) = 0.2$ ,  $p(0, 2) = 0.1$ ,  $p(1, 0) = 0.1$ ,  $p(1, 1) = 0.1$ ,  $p(1, 2) = 0.3$ . (a) Find the conditional pmf of  $Y$  given  $X=0$  and given  $X=1$ . (b) Find  $\mathbb{E}(Y | X=x)$  and  $\text{Var}(Y | X=x)$  for each  $x$ . (c) Verify the law of total variance.

**Exercise 4.24.** A cashier serves  $N$  customers, where  $N$  takes 1, 2, 3 with probabilities 0.5, 0.3, 0.2. Each customer's spend has mean 10 and variance 4. The spends are independent of each other and of  $N$ . Let  $T$  be the total. Find  $\mathbb{E}(T)$  and  $\text{Var}(T)$ .

**Exercise 4.25.** Use the joint pmf of Exercise 4.18. (a) Find  $\mathbb{E}(Y | X=0)$  and  $\mathbb{E}(Y | X=1)$ , and write down the distribution of the random variable  $\mathbb{E}(Y | X)$ . (b) Find  $\text{Var}[\mathbb{E}(Y | X)]$  and  $\mathbb{E}[\text{Var}(Y | X)]$ , and check that they add to  $\text{Var}(Y)$ . (c) What fraction of  $\text{Var}(Y)$  is the first piece? (This fraction is how much of  $Y$ 's variability  $X$  accounts for.)

**Exercise 4.26.**  $X \sim \text{Uniform}(0, 1)$ , so  $f_X(x) = 1$  on  $(0, 1)$ . Find the density of  $Y = -\ln X$ .

**Exercise 4.27.**  $X \sim \text{Uniform}(0, 1)$  and  $Y = X^3$ . Find  $f_Y$ , and compute  $\mathbb{E}(Y)$  twice: from  $f_Y$ , and without it.

**Exercise 4.28.**  $X$  and  $Y$  are independent  $\text{Uniform}(0, 1)$ . Let  $U = X$  and  $V = XY$ . (a) Find the joint density of  $(U, V)$  and the region where it is positive. (b) Integrate out  $U$  to find the density of  $V = XY$ , and check that it integrates to 1.

**Exercise 4.29.** Three letters are put at random into three addressed envelopes, one letter per envelope. Let  $X$  be the number of letters in the correct envelope. Write  $X = I_1 + I_2 + I_3$  with indicator variables and find  $\mathbb{E}(X)$  and  $\text{Var}(X)$ .

**Exercise 4.30.** Take the game of Example 4.19:  $X = 2^k$  with probability  $(\frac{1}{2})^k$  for  $k = 1, 2, 3, \dots$  (a) The bank can pay at most  $2^{10} = 1024$  dollars, so the real payoff is  $Y = X$  if  $X \leq 1024$  and  $Y = 1024$  otherwise. Find  $\mathbb{E}(Y)$ . (b) For the game with no limit, a player has  $u(x) = \sqrt{x}$ . Show that  $\mathbb{E}[u(X)]$  is finite and find it. (c) Find the sure amount  $c$  with  $u(c) = \mathbb{E}[u(X)]$ .

**Exercise 4.31.** Let  $f(x) = k(4 - x^2)$  for  $0 \leq x \leq 2$  and 0 otherwise. (a) Find  $k$ . (b) Find the CDF. (c) Find  $\mathbb{E}(X)$  and  $\text{Var}(X)$ . (d) Find  $P(X > 1 \mid X > \frac{1}{2})$ .

**Exercise 4.32.** Let  $X$  have mean  $\mu$  and variance  $\sigma^2$ . (a) Show that for every constant  $c$ ,  $\mathbb{E}[(X - c)^2] = \sigma^2 + (\mu - c)^2$ . (b) Which  $c$  makes  $\mathbb{E}[(X - c)^2]$  smallest, and what is the smallest value?

**Exercise 4.33.**  $X_1, \dots, X_n$  are i.i.d. with mean  $\mu$  and variance 4. Let  $\bar{X} = \frac{1}{n}(X_1 + \dots + X_n)$ . (a) Find  $\mathbb{E}(\bar{X})$  and  $\text{Var}(\bar{X})$ . (b) Use Chebyshev's inequality to find an  $n$  that guarantees  $P(|\bar{X} - \mu| < 0.5) \geq 0.95$ .

**Exercise 4.34.**  $X$  has the pdf  $f(x) = xe^{-x}$  for  $x > 0$  and 0 otherwise. (a) Find  $M_X(t)$  and say for which  $t$  it is finite. (b) Use it to find  $\mathbb{E}(X)$ ,  $\text{Var}(X)$  and the skewness of  $X$ . (c)  $X_1$  and  $X_2$  are independent, each with this pdf. Find the MGF, the mean and the variance of  $X_1 + X_2$ .

**Exercise 4.35.**  $X$  has the pdf  $f_X(x) = \frac{1}{4}$  on  $-1 \leq x \leq 3$  and 0 elsewhere. Let  $Y = X^2$ . (a) Find the pdf of  $Y$ . (b) Check that it integrates to 1. (c) Find  $\mathbb{E}(Y)$  twice: from the pdf of  $Y$ , and from the pdf of  $X$ .

**Exercise 4.36.** (a) Show that  $\text{Cov}(X + Y, X - Y) = \text{Var}(X) - \text{Var}(Y)$ . (b)  $X$  and  $Y$  are independent, and each is 0 or 1 with probability  $\frac{1}{2}$ . Show that  $X + Y$  and  $X - Y$  have covariance 0 but are not independent.

**Exercise 4.37.**  $X$  and  $Y$  have the joint pdf  $f(x, y) = c$  for  $0 < y < x < 2$  and 0 elsewhere. (a) Find  $c$ . (b) Find both marginal pdfs. (c) Find the conditional pdf of  $Y$  given  $X = x$ , and  $\mathbb{E}(Y \mid X = x)$ . (d) Find  $P(X + Y < 2)$ . (e) Find  $\text{Cov}(X, Y)$  and  $\rho_{XY}$ .

**Exercise 4.38.**  $X$  and  $Y$  are independent, each with the pdf  $f(x) = 1$  on  $(0, 1)$ . Let  $Z$  be the smaller of the two and  $W$  the larger. (a) Find the pdf of  $W$  and the pdf of  $Z$  with the CDF method. (b) Find  $\mathbb{E}(Z)$  and  $\mathbb{E}(W)$ . (c) Find  $\text{Cov}(Z, W)$ .

**Exercise 4.39.**  $X$  has the pdf  $f_X(x) = 2x$  on  $(0, 1)$ . Given  $X = x$ ,  $Y$  has the pdf  $\frac{1}{x}$  on  $(0, x)$ . (a) Find the joint pdf and its region. (b) Find the marginal pdf of  $Y$ . (c) Find  $P(Y < \frac{1}{2} \mid X = \frac{3}{4})$  and  $P(X > \frac{1}{2} \mid Y = \frac{1}{4})$ . (d) Find  $\mathbb{E}(Y)$  with the law of iterated expectations and  $\text{Var}(Y)$  with the law of total variance. Check both from the pdf of  $Y$ .

**Exercise 4.40.**  $X$  and  $Y$  have the joint pdf  $f(x, y) = e^{-x}$  for  $0 < y < x$  and 0 elsewhere. (a) Find both marginal pdfs. (b) Find  $P(X < 3, Y > 1)$ . (c) Find  $P(X - Y \geq 1)$ . (d) Find the conditional pdf of  $X$  given  $Y = y$  and  $\mathbb{E}(X \mid Y = y)$ . Then check the law of iterated expectations.

**Exercise 4.41.** The returns  $R_1, \dots, R_n$  of  $n$  assets each have variance  $\sigma^2$ , and every pair has covariance  $c$ . A portfolio puts weight  $\frac{1}{n}$  on each asset. Its return is  $R_p = \frac{1}{n}(R_1 + \dots + R_n)$ . (a) Show that  $\text{Var}(R_p) = \frac{\sigma^2}{n} + \frac{n-1}{n}c$ . (b) What happens as  $n$  grows? Say in one sentence what diversification can and cannot remove.

**Exercise 4.42.** Let  $X$  and  $Y$  have positive variances. For a real number  $t$  put  $h(t) = \text{Var}(X - tY)$ . (a) Write  $h(t)$  as a quadratic in  $t$ . (b) Use  $h(t) \geq 0$  for every  $t$  to show that  $[\text{Cov}(X, Y)]^2 \leq \text{Var}(X) \text{Var}(Y)$ .

(c) What does (b) say about  $\rho_{XY}$ ?

**Exercise 4.43.**  $X$  takes the values  $0, 1, 2, \dots$  and has a finite mean. (a) Show that  $\mathbb{E}(X) = \sum_{k=1}^{\infty} P(X \geq k)$ . (b) Two fair dice are rolled, and  $X$  is the smaller of the two faces. Find  $P(X \geq k)$  for  $k = 1, \dots, 6$ , and use (a) to find  $\mathbb{E}(X)$ .

**Exercise 4.44.** A fair coin is tossed until two heads appear in a row. Let  $N$  be the number of tosses. Condition on the first tosses to find  $\mathbb{E}(N)$ . You may assume that  $\mathbb{E}(N)$  is finite.

**Exercise 4.45.**  $X$  and  $Y$  are independent, each with the pdf  $e^{-x}$  on  $(0, \infty)$ . Let  $U = X+Y$  and  $V = \frac{X}{X+Y}$ . (a) Find the joint pdf of  $(U, V)$  and its region. (b) Find the pdf of  $U$  and the pdf of  $V$ . (c) Are  $U$  and  $V$  independent?